

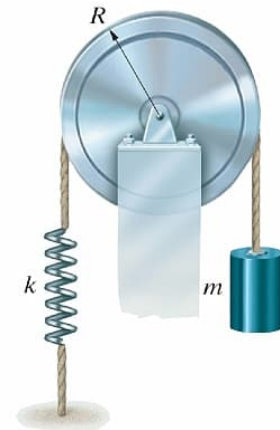
Focus of this homework:

Check the basic technical background and preparedness of students in dynamics and ordinary differential equations.

Submit homework (in printed version) by the beginning of the class. It is your responsibility to make sure that homework is submitted in good quality. Please organize your work neatly. Suggest to start each problem on a new page.

- 0) (Optional) In less than one page (but at least one paragraph), introduce yourself to the instructor such as your academic major, your expectation of this class, etc. Please compose in English. Attach one photo of yourself (optional). Email this to the instructor (tan@wayne.edu), i.e., submit separately from the homework.
The email subject heading should be: "2026 ZJU/ZJUT Vibration class – Your name".

- 1) The uniform pulley has radius R and mass moment of inertia I_O about its center of mass at O . Assume that the spring is linear, i.e., the force in the spring $F_s = k\Delta$, where Δ is the deflection of the spring from its unstretched configuration (natural length).
- a) Draw the free-body diagrams and derive the governing equation of motion for this system using an absolute coordinate x . The coordinate x measures the downward displacement of the mass relative to its position when the spring is unstretched. [Dynamics]
- b) Suppose the mass is released from rest when the spring is unstretched, solve the governing differential equation analytically. (Remember to properly set the initial conditions.) [Differential equation]
- c) Choose your own values of the parameters (justify your choice of the parametric values) or use the following suggested values to simulate the response of this system. Plot your analytical solution against the simulation result and compare. You may use MATLAB or any other suitable software. [Numerical simulation]



To simulate the response, the following values for the parameters are selected. These values are selected based on an athlete using such a prototypical training machine in a gym. Note that the static deflection is about 2 cm.

$$m = 100 \text{ kg}, \quad k = 50,000 \text{ N/m}, \quad R = 0.15 \text{ m}, \quad I_O = 2 \text{ kg}\cdot\text{m}^2$$

- d) Write down the key observations of your solutions. What have you learned from this problem? [Observation of physics]